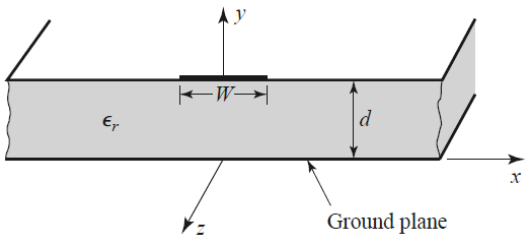
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EXERCISE SESSION 3 (20.11.2020)

Problem 3.1: Microstrip line

The following are the design formulations for microstrip line:



$$Z_0 = \begin{cases} \dfrac{60}{\sqrt{\epsilon_e}} \ln\left(\dfrac{8d}{W} + \dfrac{W}{4d}\right) & \text{for } W/d \le 1 \\ \dfrac{120\pi}{\sqrt{\epsilon_e}\,[W/d + 1.393 + 0.667 \ln(W/d + 1.444)]} & \text{for } W/d \ge 1. \end{cases}$$

where, ε_e is the effective dielectric constant of the microstrip line given by

$$\epsilon_e = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \frac{1}{\sqrt{1 + 12d/W}}$$

The dielectric loss from the substrate is given by

$$\alpha_d = \frac{k_0 \epsilon_r (\epsilon_e - 1) \tan\delta}{2\sqrt{\epsilon_e}(\epsilon_r - 1)} \text{ Np/m}$$

The conduction loss is given by

$$\alpha_c = \frac{R_s}{Z_0 W} \text{ Np/m}$$

Where, R_s is the surface resistivity of the conductor and can be calculated as

$$R_s = \sqrt{\omega\mu_0/2\sigma}$$

A microstrip circuit design operating at 100 GHz is realized on a substrate of dielectric constant 3, loss tangent 10^{-4} and of thickness 2.5 mm. The metal conductor has a conductivity of 10^8 S/m. It consists of three different lines 'M1' (width 2 mm, length 5 mm), 'M2' (width 1 mm, length 5 mm) and 'M3' (width 0.5 mm, length 5 mm) and is left open as shown in figure below. Due to fringing field on the open end, a capacitance of 10 pF is generated at the operational frequency.

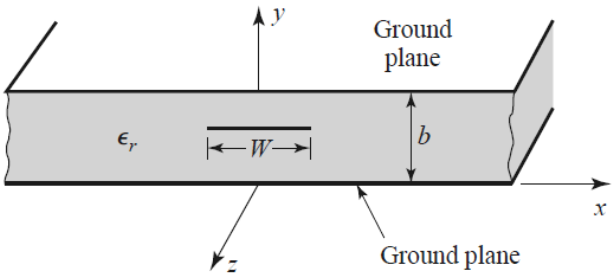


Calculate the magnitude (in dB) and phase (in degrees) of input reflection coefficient at port 1, which has an impedance of 50 Ω.

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Problem 3.2: Stripline

The following are the design formulations for stripline:

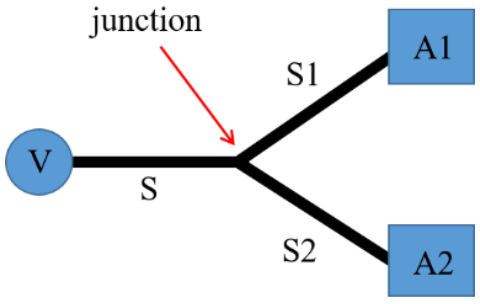


$$Z_0 = \frac{30\pi}{\sqrt{\epsilon_r}} \frac{b}{W_e + 0.441b}$$

where, *We* is the effective width of the center conductor given by

$$\frac{W_e}{b} = \frac{W}{b} - \begin{cases} 0 & \text{for } \dfrac{W}{b} > 0.35 \\ (0.35 - W/b)^2 & \text{for } \dfrac{W}{b} < 0.35. \end{cases}$$

As shown in the figure below, two different antennas 'A1' with input impedance 50 Ω and 'A2' with input impedance 65 Ω, both operating at 2.4 GHz are fed by two different striplines 'S1' and 'S2', respectively. Both the striplines are connected to the source via a 3-port junction, which is to be assumed perfect and lossless. The whole circuit is designed in a planar substrate of thickness 2 mm, dielectric constant 3, shows no dielectric dissipation loss and no conduction loss.



a) Find out the width of striplines 'S1' and 'S2', if they are matched perfectly to the antenna.
b) Assume that both the antennas need to be fed with equal power. If the length of stripline 'S1' from the junction to the antenna 'A1' is 25 mm, find the corresponding length of the stripline 'S2' from the junction to the antenna 'A2'.
c) In the above scenario c), calculate width of the stripline 'S' connecting the source and junction, if a perfect matched junction from the input side needs to be designed.